

Telegram



Discord



JS → programming language → high lvl vs low level
↓
human readable binary
0 & 1

↳ why JS?

↳ frontend, backend, JS → Browser

where JS run

↳ Browser, html with script tag
→ ME → node.js

variables in JS

↓
box / container

↓
value

age = 15
↓ ↓
box value

3 keywords

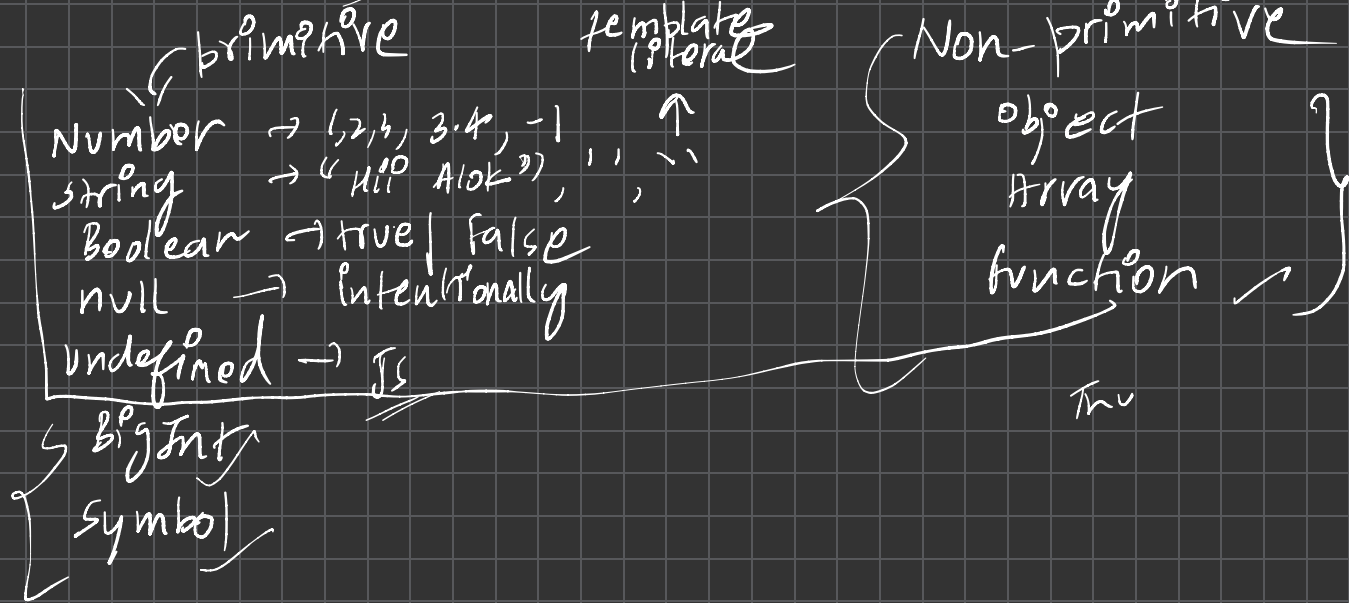
var, let & const

↳ redclaration
update

↓
update

↓
no-redclaration
no update

Data types



operators

Symbol $\rightarrow$ do operands ke
beech operation perform krta
hai

$\rightarrow$ Arithmetic operators
(Maths)

$+$, $-$, $*$, $/$, $\%$, $**$

$\downarrow$ addition $\downarrow$ multiply $\downarrow$ division (quotient) $\downarrow$ modulus (remainder) $\rightarrow$ exponentiation (power)

subtraction

2^3

```
let num1 = 2;
```

```
let num2 = 4;
```

```
console.log(num1 + num2);
```

```
console.log(num1 - num2);
```

```
console.log(num1 * num2);
```

```
console.log(num2 / num1);
```

```
console.log(num2 % num1);
```

```
console.log(num1 ** num2);
```

6 ✓

-2 ✓

8 ✓

2 ✓

0 ✓

16

left right operand

$2^4 \rightarrow 2 \times 2 \times 2 \times 2$

$2 \overline{) 361}$
 $\underline{2}$
 $\underline{1}$ ✓

dividend
divisor
quotient
remainder

$2 \overline{) 42}$
 $\underline{4}$
 $\underline{0}$ ✓

$\div$ → divide / quotient

$\% \rightarrow$ modulus
remainder

JS case sensitive

```
// case sensitive  
// teeno variable different hai  
  
let NAME = "saif"  
let name = "juhi"  
let Name = "karan"
```

(++) increment & decrement (--)
↳ post
↳ pre

+=, -

num++
↳ num = num + 1;

num++;

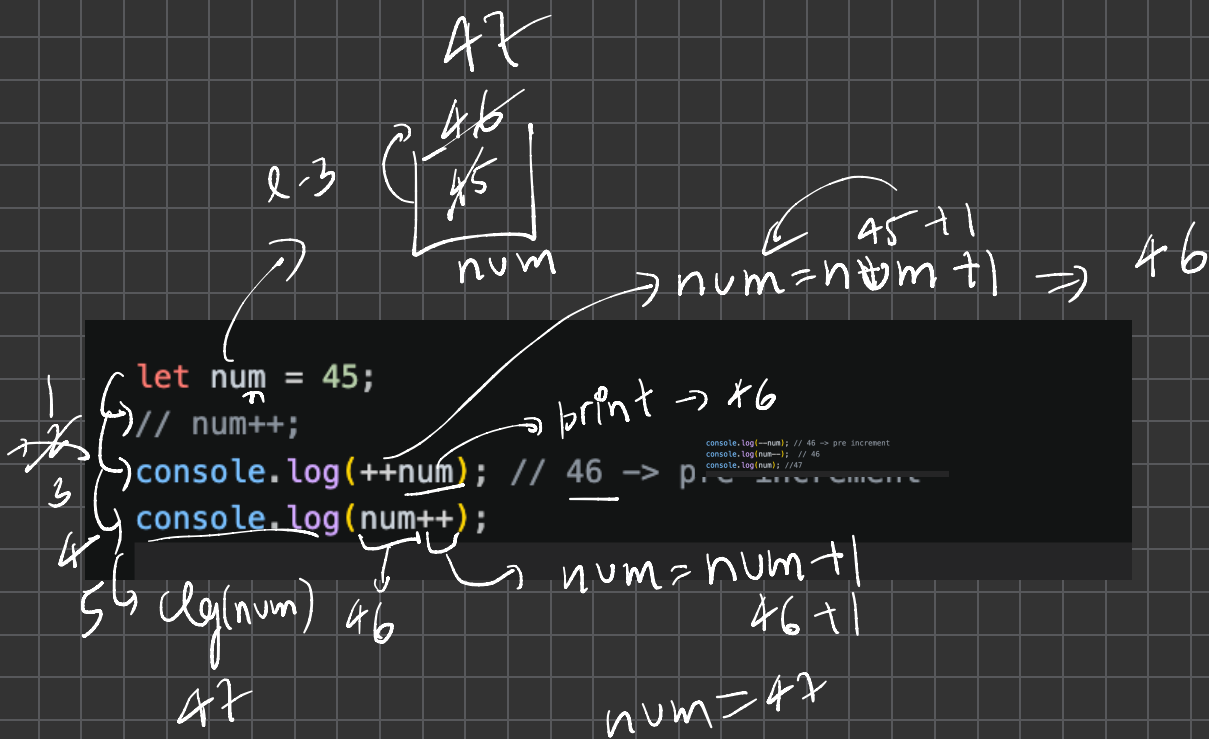
```
let num = 45; ✓  
num++;  
console.log(num);  
|
```

```
let num = 45; // num++;  
console.log(++num); // 46 -> pre increment  
console.log(num++);
```

Handwritten annotations: A bracket on the left groups the two log statements. An arrow points from the first log statement to the value 46. An arrow points from the second log statement to the value 46.

```
let num = 45; ✓  
// num++;  
// console.log(++num); // 46 -> pre increment  
console.log(num++);
```

Handwritten annotations: A bracket on the left groups the last two lines. An arrow points from the last line to the value 46. An arrow points from the last line to the value 45. A large arrow points from the last line to the text log(num) -> 46.



num = ~~45~~ ~~44~~ 43

```
console.log(--num); // 40 → pre increment  
console.log(num--); // 41  
console.log(num); // 42
```

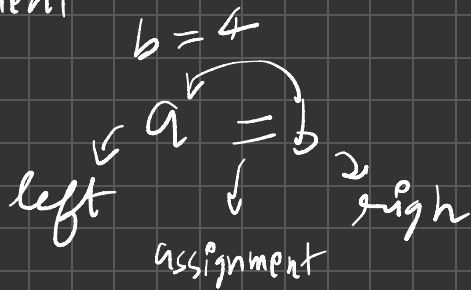
↓
43

num--
↳ num = num - 1

Assignment operator

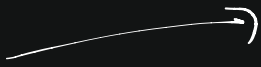
=, +=, -=, *=, /=, %=, **=

assignment



$a = 4$

```
✓ let num = 2;  
  num += 5;  
  console.log(num);
```



$num = num + 5;$

$num += 5$

$num = +5$



$num = +5$

```

let num = 2;
num += 5;    → num = num + 5
console.log(num);    → 7
num -= 5;    → num = num - 5
console.log(num);    → 2
num *= 5;    → num = num * 5
console.log(num);    → 10

```

7 2 10

```

num /= 5;    → num = num / 5
console.log(num);    → 2
num %= 5;    → num = num % 5
console.log(num);    → 2
num **= 5;   → num = num ** 5
console.log(num);    → 32

```

7 2 10 2 2 32
2 % 5 = 2

```
let num = 2;  
num += 5;  
console.log(num);  
num -= 5;  
console.log(num);  
num *= 5;  
console.log(num);  
num /= 5;  
console.log(num);  
num %=5;  
console.log(num);  
num **=5 ;  
console.log(num);
```

comparison operator $\rightarrow$ always return boolean

$>$ $\rightarrow$ greater than

$<$ $\rightarrow$ less than

$>=$ $\rightarrow$ greater than or equal

$<=$ $\rightarrow$ less than or equal

$=$ $\rightarrow$ loose equality

$===$ $\rightarrow$ strict equality

$!=$ $\rightarrow$ not equals to

```
const num1 = 3;  
const num2 = 6;
```

```
console.log(3 > 6)  
console.log(3 < 6)  
console.log(3 >= 6)  
console.log(3 <= 6)  
console.log(3 == 6)  
console.log(3 != 6)
```

false
T
F
T
F
T

F T F T F T

3 $\neq$ 5
0 $\neq$

$b > b$	F
$b < b$	F
$b == b$	T
$b >= b$	T
$b <= b$	T

Value must same

→ loose equality
(==)

→ strict equality
(===)

type check nhi krta

```
console.log("5" == 5);
```

humesha type check krta hai and type must be same

```
// loose equality  
console.log("5" == 5);
```

```
// strict equality  
console.log("5" === 5);
```

true
false

Logical operators

&& → AND → dono true hone chahiye

|| → OR → ek bhi true hoga chalega

! → NOT → opposite krdo

Truth Table

<u><u>A</u></u>	<u>B</u>	
True	True	$\rightarrow$ True ✓
True	False	$\rightarrow$ false
False	True	$\rightarrow$ false
False	False	$\rightarrow$ false

club

$18 \geq$
Id ✓

```
const age = 18;
const hasId = true;
const canEnterClub = (age >= 18 && hasId === true);
console.log(canEnterClub);
```

online
vbi ✓
cards X

OR Truth Table

<u>A</u>	<u>B</u>	
True	True	$\rightarrow$ True
True	False	$\rightarrow$ True
False	True	$\rightarrow$ True
false	False	$\rightarrow$ false ✓

! $\rightarrow$ opposite

Truthy & falsy value

falsy

- 1 0 → zero
- 2 "" → empty string
- 3 null →
- 4 undefined
- 5 NaN
- 6 false

Truthy

"hi"

5666

true

True $\rightarrow 1$
False $\rightarrow 0$

Operator precedence
BODMAS

$$2 + 3 \times 4$$

Conditionals

↳ condition

Bank Account / SPM card

18+

If else

condition

↳ boolean

if {

} {

} else {

↳ no condition

}

youtube

↳ logged in

↳ True

↳ like, comment ✓

↳ not logged in

↳ like X, comment X

False
if (true) {



→ True block

} else {



✓ false block

}

if / else if / else

key

fri === mon → false

fri === Tue

①

```
let day = "fri";  
if (day === "mon") {  
  console.log("1st day of the week");  
} else if (day === "tue") {  
  console.log("2nd day of the week");  
} else if (day === "wed") {  
  console.log("3rd day of the week");  
} else if (day === "thu") {  
  console.log("4th day of the week");  
} else if (day === "fri") {  
  console.log("5th day of the week");  
} else if (day === "sat") {  
  console.log("6th day of the week");  
} else if (day === "sun") {  
  console.log("7th day of the week");  
} else {  
  console.log("Wrong day");  
}
```

90 → A
80-90 → B
70-80 → C
60-70 → D
≤ 60 → F

Nested if else

if logged in
↳ if subscribed ✓

else →

else

→ please

↳ Netflix

↳ subscription

↳ free user

code readability

Switch case

Switch case

switch(expression) {

case value1:

==>

break;

case value2:

==>

break;

default:

==>

}

day = "fri" ?
cases

```
switch (day) {  
  case "mon":  
    console.log("1st day of the week");  
    break;  
  case "tue":  
    console.log("2nd day of the week");  
    break;  
}
```

```
const day = "fri";
```

```
switch (day) {
```

```
  ✓ case "mon":  
    console.log("1st day of the week");
```

```
  ✓ case "tue":  
    console.log("2nd day of the week");
```

```
  ✓ case "wed":  
    console.log("3rd day of the week");
```

```
  ✓ case "thu":  
    console.log("4th day of the week");
```

```
  ✓ case "fri":  
    console.log("5th day of the week"); → print
```

```
  case "sat":  
    console.log("6th day of the week"); → print
```

```
  case "sun":  
    console.log("7th day of the week"); → print
```

```
  default:  
    console.log("Wrong day"); → print
```

```
}
```

→ break;